

Human Natural Structure (HNS)

Foundational Specification

Version 1.1

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Revision Notice — Version 1.1

This version corrects three structural inconsistencies identified in HNS 1.0:

- (1)** Section 7.3 and 8.3: The Layer and Category ordering sequences, previously referenced as figures, are now fully specified in text as normative content.
- (2)** Section 10: The Recomposition step, described in Section 6.4 for composite phenomena, has been added to the Mapping Procedure (Section 10.1, Step 6) for normative consistency.
- (3)** Minor cross-reference clarifications throughout.

Purpose Declaration

The purpose of this volume is to formally define the foundational coordinate system (the 36-Matrix) that underlies the entire Human Natural Structure (HNS) and thereby establish the structural basis of HNS 1.0.

This specification does not introduce new theories or additional content.

Instead, it serves as the final foundational specification that unifies the six existing volumes and the broader HNS framework under a single, coherent coordinate system.

The 36-Matrix provides the standardized grid through which phenomena relevant to human, social, or civilizational structure can be positioned at the intersection of six Natural Layers (causal structure) x six Categories (non-causal explanatory axes), establishing the reference framework for all future applications, analyses, and interpretations of HNS.

From Description to Coordinates

The six existing volumes of the HNS Series represent an extensive effort to build descriptive accounts of human, social, and civilizational structure. By detailing the body, cognition, interaction, environment, load, and patterns, the series has created a rich topography of human functioning.

Yet description alone cannot determine where a new or unknown phenomenon should be placed.

Description provides depth, but not position.

The HNS 36-Matrix defined in this volume addresses this limitation by introducing a coordinate system. By combining the six Human Natural Layers (causal structure) with the six Abstract Cognitive Categories (non-causal explanatory axes), phenomena relevant to human, social, or civilizational structure can be positioned at the intersection of L x C.

From description to coordinates. From terrain maps to a geodetic system.

This volume provides the foundational coordinate framework that unifies and interprets the entire HNS Series.

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1. Introduction

The Human Natural Structure (HNS) is a foundational coordinate system designed to describe the causal architecture and explanatory dimensions of human, social, and civilizational phenomena. This specification defines the structural components, interpretive rules, and standardized formats that constitute HNS 1.0.

HNS provides a unified framework for positioning any phenomenon within a 36-cell matrix, formed by the intersection of:

- six Human Natural Layers (causal strata), and
- six Abstract Cognitive Categories (explanatory axes).

This document establishes the formal specification of that framework.

1.1 Purpose of This Specification

The purpose of the HNS 1.0 Foundational Specification is to:

- define the structural architecture of HNS,
- standardize the format and interpretation of all 36 cells,
- provide a consistent method for mapping phenomena,
- ensure interoperability across disciplines and applications,
- and establish a stable reference for future research and extensions.

This specification is normative, except where explicitly noted as non-normative.

1.2 Scope

This document covers:

- the definition of Human Natural Layers (Section 7),
- the definition of Abstract Cognitive Categories (Section 8),
- the structural specification of the 36-Matrix (Section 5),
- the standardized Cell Format (Section 2),
- the 36-Cell Template (Section 9),
- the Mapping Rules (Section 10),
- and the authoritative Definition Statements (Section 11).

It does not prescribe domain-specific interpretations or applications. HNS is domain-agnostic and may be used in any field involving human, social, or civilizational structure.

1.3 Structure of This Document

This specification is organized as follows. Sections 3, 4, and 6 are informative. All other sections are normative.

1.4 Intended Audience

This specification is intended for researchers and analysts, organizational designers, cognitive and social scientists, institutional architects, systems theorists, and any practitioner working with human or civilizational structure. No prior knowledge is required beyond familiarity with structural analysis.

1.5 Reading HNS

Readers should understand the following principles before using HNS:

- Layers are causal. They describe where a phenomenon originates.
- Categories are explanatory. They describe how a phenomenon is interpreted.
- Cells are structural positions. They do not prescribe content; they define coordinates.
- Mapping is rule-based. All mappings must follow the procedures in Section 10.
- The 36-Matrix is a coordinate system, not a taxonomy. It positions phenomena; it does not classify them by type.

2. Cell Format

The Cell Format defines the mandatory structural components that constitute any cell within the HNS 36-Matrix. A cell represents the intersection of one Human Natural Layer (L_i) and one Abstract Cognitive Category (C_j), and must follow a standardized internal structure to ensure consistency, interpretability, and interoperability across the entire HNS framework.

This section defines the format, not the content, of each cell.

2.1 Purpose of the Cell Format

The Cell Format serves three primary purposes:

- **Standardization** — Ensures that all 36 cells follow a unified structural pattern.
- **Interpretive Clarity** — Provides a consistent internal structure that supports accurate mapping and analysis.
- **Specification Integrity** — Establishes the foundational structure upon which the 36-Matrix is built.

2.2 Mandatory Components of a Cell

Each cell must contain the following components:

1. **Cell Identifier** — The formal notation representing the Layer-Category intersection.
2. **Layer Reference** — A direct reference to the Human Natural Layer (Section 7) that defines the causal domain.
3. **Category Reference** — A direct reference to the Abstract Cognitive Category (Section 8) that defines the explanatory axis.
4. **Structural Role Statement** — A formal description of the structural function of this intersection.
5. **Interpretive Scope** — The range of phenomena that may be positioned within this cell.
6. **Boundary Conditions** — Rules for distinguishing this cell from adjacent cells.
7. **Misclassification Prevention** — Guidelines to avoid interpretive errors.
8. **Optional Non-normative Examples** — Illustrative examples that clarify the structural meaning of the cell.

These components define the minimum required structure for any cell.

2.3 Structural Constraints

- **Uniqueness** — Each cell represents a unique structural position.
- **Non-overlap** — No two cells may share the same structural domain.
- **Bidirectional Independence** — The meaning of a cell cannot be derived from the Layer or Category alone; it emerges from their intersection.
- **Interpretive Neutrality** — Cells define structural positions, not prescriptive content.
- **Template Compliance** — All cells must conform to the template defined in Section 9.

2.4 Internal Ordering Requirements

Within each cell, components must appear in the following order: (1) Cell Identifier, (2) Layer Reference, (3) Category Reference, (4) Structural Role, (5) Interpretive Scope, (6) Boundary Conditions, (7) Misclassification Prevention, (8) Optional Examples. This ordering ensures consistency across all 36 cells.

2.5 Relationship to the 36-Matrix

Section 2 defines what a cell is. Section 5 defines how cells exist as structural positions. Section 9 defines how cells are documented. Together, these sections form the complete specification of the HNS coordinate system.

3. Layer-by-Layer [Informative]

The Human Natural Layers represent the causal architecture of HNS. This section provides an explanatory overview of each layer, describing how the six strata relate to one another and how they collectively form the vertical axis of the 36-Matrix. While Section 7 defines the layers formally, this section explains how they function, interact, and build upon one another.

3.1 Layer 1 — PhysicalOS

Physical and biological foundations

Layer 1 forms the base substrate of human existence. It includes biological systems, neural structures, physical constraints, and material conditions. All higher layers depend on the stability and functioning of this physical foundation. Without PhysicalOS, no cognition, interaction, or emergent pattern can occur.

3.2 Layer 2 — CognitiveOS

Internal processing and representation

Layer 2 transforms physical signals into structured cognition. It includes perception, memory, internal processing, interpretation, and decision-making. CognitiveOS depends on PhysicalOS, and in turn enables the intentional behaviors that appear in Layer 3.

3.3 Layer 3 — InteractionOS

Behavioral exchange and interaction

Layer 3 represents observable interaction: communication, cooperation, conflict, behavioral exchange, and human-environment interaction. InteractionOS is where cognition becomes action — the layer where individuals affect one another and their surroundings.

3.4 Layer 4 — EnvironmentOS

Structures surrounding human activity

Layer 4 consists of the external structures that shape human behavior: physical environments, social environments, institutional systems, and informational environments. EnvironmentOS provides the context and constraints within which interactions occur. It is not behavior itself, but the structure that surrounds behavior.

3.5 Layer 5 — LoadOS

Pressures, demands, and constraints

Layer 5 represents the loads acting on individuals or systems: internal loads (stress, fatigue, cognitive load), external loads (demands, scarcity, pressure), and systemic loads (resource constraints, institutional burdens). LoadOS determines capacity, strain, and adaptive response across all layers.

3.6 Layer 6 — PatternOS

Emergent outcomes and long-term regularities

Layer 6 captures emergent patterns: cultural norms, behavioral regularities, long-term trends, and systemic outcomes. PatternOS is not a cause but an emergent result of interactions across all lower layers. It represents the long-term structure of human and civilizational functioning.

4. Category Integration [Informative]

The Abstract Cognitive Categories provide the non-causal explanatory axes of HNS. While the Human Natural Layers define where a phenomenon belongs in the causal architecture, the Categories define how it can be interpreted, analyzed, or conceptualized.

4.1 Civilization Categories

Macro-scale, historical, and civilizational framing

Civilization Categories provide the broadest interpretive lens. They frame phenomena in terms of civilizational structures, historical transitions, macro-level dynamics, and epochal patterns. This Category is used when the goal is to understand a phenomenon at the scale of societies, civilizations, or long-term historical processes.

4.2 Core Categories

Foundational principles and essential structures

Core Categories identify the fundamental logic of a phenomenon. They focus on essential structures, irreducible principles, foundational mechanisms, and conceptual kernels. This Category is used when the goal is to clarify: what is the essential structure or principle behind this phenomenon?

4.3 Module Categories

Components, elements, and functional units

Module Categories break phenomena into discrete, analyzable units: components, elements, sub-systems, and functional parts. This Category is used when the goal is to understand: what are the parts that make up this phenomenon?

4.4 Application Categories

Practical uses and applied contexts

Application Categories interpret phenomena in terms of practical use, applied function, operational context, and real-world implementation. This Category is used when the goal is to understand: how is this phenomenon used or applied?

4.5 System Categories

Organized mechanisms and structured processes

System Categories explain phenomena as part of organized systems, mechanisms, institutional processes, and structured operations. This Category is used when the goal is to understand: how does this phenomenon operate as a system?

4.6 External Categories

Outer contextual factors and external influences

External Categories frame phenomena in terms of external conditions, contextual influences, surrounding constraints, and outer environmental factors. This Category is used when the goal is to understand: what external factors shape or influence this phenomenon?

5. 36-Cell Specification

The 36-Cell Specification defines the structural properties, constraints, and interpretive requirements of each of the 36 positions created by the intersection of six Human Natural Layers (L1-L6) and six Abstract Cognitive Categories (C1-C6). This section does not define the descriptive content of each cell. Instead, it establishes the formal specification that governs how each cell exists, how it must be interpreted, and how it relates to the other cells within the HNS 36-Matrix.

5.1 Structural Identity of Each Cell

Each cell in the 36-Matrix is defined by a unique intersection of one Human Natural Layer (L_i) and one Abstract Cognitive Category (C_j), written as Cell ($L_i \times C_j$). Each cell represents a distinct structural domain that cannot be reduced, merged, or substituted.

5.2 Mandatory Cell Properties

- **Uniqueness** — Each cell represents a unique structural position that cannot be duplicated.
- **Non-overlap** — No two cells may share the same structural domain.
- **Bidirectional Independence** — The Layer and Category axes remain independent; the meaning of a cell cannot be derived from one axis alone.
- **Interpretive Neutrality** — Cells define structural positions, not prescriptive content.
- **Template Compliance** — Each cell must follow the standardized format defined in Section 9.

5.3 Internal Structure of a Cell

Each cell must contain: (1) Layer Reference, (2) Category Reference, (3) Structural Role Statement, (4) Interpretive Scope, (5) Boundary Conditions, (6) Misclassification Prevention, and (7) Optional Non-normative Examples. These components define the minimum required structure.

5.4 Matrix-Level Constraints

- **Exhaustiveness** — All possible Layer-Category intersections must be represented.
- **Mutual Exclusivity** — No phenomenon may occupy more than one cell simultaneously.
- **Structural Completeness** — The 36 cells collectively define the full coordinate space of HNS.
- **Interpretive Stability** — The structural meaning of each cell must remain stable across contexts.

5.5 Cell Adjacency Rules

Cells may be adjacent in two ways:

- **Horizontal adjacency** — Same Layer, Different Category: $L_i \times C_j \rightarrow L_i \times C_{(j+1)}$
- **Vertical adjacency** — Same Category, Different Layer: $L_i \times C_j \rightarrow L_{(i+1)} \times C_j$

Adjacency does not imply similarity. Boundary conditions must be used to distinguish adjacent cells.

5.6 Cell Interpretation Rules

- **Layer-First Interpretation** — The causal domain (Layer) determines the primary meaning.
- **Category-Second Interpretation** — The explanatory purpose (Category) refines the interpretation.
- **No Cross-Cell Inference** — The meaning of one cell cannot be inferred from another.

- **No Diagonal Interpretation** — Diagonal relationships have no structural meaning.

6. Usage Notes [Non-normative]

This section provides non-normative notes intended to support the practical use and interpretation of HNS. These notes do not modify the formal specification.

6.1 Understanding Layers vs. Categories

HNS is built on two independent axes: Human Natural Layers (causal structure) and Abstract Cognitive Categories (explanatory structure). Layers answer 'Where does this phenomenon originate causally?' Categories answer 'How are we interpreting this phenomenon?' Keeping these roles distinct is essential for correct mapping.

6.2 Interpreting the 36-Matrix

The 36-Matrix is a coordinate system, not a taxonomy. It positions phenomena by causal origin x explanatory purpose. A phenomenon can appear in different cells depending on the interpretive purpose. The same phenomenon may be decomposed into multiple components across different cells. No cell is 'higher' or 'lower' in importance; the matrix is structurally neutral.

6.3 Working with Composite Phenomena

Many real-world phenomena span multiple layers or categories. The recommended approach is:

- **Decomposition** — Break the phenomenon into causal components.
- **Primary assignment** — Map each component to its correct cell.
- **Recomposition** — Understand the phenomenon as a structured whole by reading the mapped components together.

Note: Recomposition is an interpretive step; it does not assign a phenomenon to a new cell. Each component retains its individual cell assignment. For normative mapping rules on composite phenomena, see Section 10.5.

6.4 Typical Use Cases

HNS can be applied in: analysis of human behavior, organizational diagnosis, institutional design, cultural interpretation, cognitive modeling, systemic analysis, and civilizational studies. HNS is domain-agnostic and can be used wherever human, social, or civilizational structure is relevant.

6.5 Common Misunderstandings

- **Confusing Patterns with Causes** — PatternOS is emergent, not causal.
- **Treating Categories as hierarchical** — Categories are interpretive lenses, not levels.
- **Assigning based on scale or importance** — Neither determines Layer or Category.
- **Assuming diagonal relationships** — Diagonal cells have no structural meaning.
- **Overusing External Categories** — External does not mean 'anything outside'; it refers to structured external influences.

7. Human Natural Layers [Normative]

The Human Natural Layers constitute the six causal strata that define the natural structure of humans, societies, and civilizations. These layers represent the irreducible causal architecture of HNS. Each layer is distinct, non-overlapping, and vertically ordered, forming a causal hierarchy from physical foundations to emergent patterns. The Human Natural Layers serve as the primary axis (L) of the HNS 36-Matrix.

7.1 Layer Structure Requirements

- **Causality** — The layer must represent a causal domain that produces effects in the layers above it.
- **Non-redundancy** — No layer may duplicate the causal function of another layer.
- **Completeness** — The six layers collectively exhaust the causal structure of human and civilizational functioning.
- **Vertical Ordering** — Layers must be interpreted from foundational (Layer 1) to emergent (Layer 6).
- **Single Assignment** — A phenomenon must be assigned to exactly one Human Natural Layer.

7.2 Formal Definitions of the Six Layers

Layer 1 — PhysicalOS

Definition: The physical and biological foundations that enable all higher-order functions. Includes anatomy, physiology, neural substrates, and material constraints.

Causal Role: Provides the substrate upon which cognition, interaction, and all emergent patterns depend.

Layer 2 — CognitiveOS

Definition: Internal cognitive processes including perception, memory, processing, and internal representation.

Causal Role: Transforms physical signals into structured cognition and enables decision-making, interpretation, and internal modeling.

Layer 3 — InteractionOS

Definition: Human-to-human and human-to-environment interactions, including communication, coordination, and behavioral exchange.

Causal Role: Generates social and environmental effects through observable actions and interactions.

Layer 4 — EnvironmentOS

Definition: The physical, social, institutional, and informational environments within which interactions occur.

Causal Role: Provides the external structures that shape and constrain human behavior and social dynamics.

Layer 5 — LoadOS

Definition: Internal and external loads, pressures, constraints, and demands acting upon individuals or systems.

Causal Role: Determines stress, capacity, resource allocation, and adaptive responses across layers.

Layer 6 — PatternOS

Definition: Behavioral patterns, emergent outcomes, and long-term regularities produced by interactions across all lower layers.

Causal Role: Represents the emergent structure of human and civilizational functioning.

7.3 Layer Ordering Principle [REVISED in v1.1]

v1.1 Revision: The ordering sequence previously referenced as a figure is now fully specified in text as normative content.

The Human Natural Layers must always be interpreted in ascending causal order, from foundational to emergent. The mandatory sequence is:

Position	Layer	Causal Role
L1 (first)	PhysicalOS	Physical and biological substrate
L2	CognitiveOS	Internal cognitive processing
L3	InteractionOS	Behavioral exchange and interaction
L4	EnvironmentOS	Surrounding structures and contexts
L5	LoadOS	Pressures, demands, and constraints
L6 (last)	PatternOS	Emergent outcomes and regularities

This ordering is mandatory for all mapping, interpretation, and analysis within HNS. Layer 1 always precedes Layer 2; Layer 2 always precedes Layer 3; and so forth through Layer 6. No layer may be interpreted before its causal predecessors.

7.4 Layer Assignment Rule

A phenomenon must be assigned to a Human Natural Layer based on its primary causal domain, not its appearance, context, or effects. Example: a social behavior belongs to InteractionOS, even if influenced by cognition or environment.

7.5 Boundary Conditions

- **Primary Causality Rule** — Assign to the layer that provides its primary causal mechanism.
- **Effect vs. Origin Distinction** — Effects may appear in higher layers, but origin determines assignment.
- **Composite Phenomena Rule** — A composite phenomenon must be decomposed into its causal components, each assigned separately.
- **Environmental Influence Exception** — Environmental factors may influence cognition or interaction, but remain in EnvironmentOS unless they directly constitute behavior or cognition.
- **Emergent Pattern Clarification** — Patterns belong to PatternOS, even if arising from repeated interactions or cognitive processes.

7.6 Misclassification Prevention

- **PatternOS as Cause** — Patterns are emergent outcomes, not causal origins.
- **LoadOS vs. EnvironmentOS** — Loads are pressures acting on a system; environments are structures surrounding it.

- **InteractionOS vs. PatternOS** — Interaction is event-level; patterns are aggregated outcomes.
- **CognitiveOS vs. InteractionOS** — Internal processing belongs to cognition; observable exchange belongs to interaction.
- **PhysicalOS vs. CognitiveOS** — Neural substrates are physical; mental processes are cognitive.

7.7 Examples [Non-normative]

- PhysicalOS: reflexes, muscle fatigue, neural firing
- CognitiveOS: memory retrieval, perception, decision-making
- InteractionOS: conversation, cooperation, conflict
- EnvironmentOS: laws, institutions, physical settings
- LoadOS: stress, workload, resource scarcity
- PatternOS: culture, norms, long-term behavioral trends

8. Abstract Cognitive Categories [Normative]

The Abstract Cognitive Categories constitute the six non-causal explanatory axes that organize interpretation across the Human Natural Layers. While the Layers define where a phenomenon belongs in the causal structure, the Categories define how it is interpreted, analyzed, or conceptualized. The Categories serve as the horizontal axis (C) of the HNS 36-Matrix.

8.1 Category Structure Requirements

- **Non-causality** — Categories must not represent causal domains; they provide explanatory perspectives.
- **Cross-layer Applicability** — Each Category must apply to all six Human Natural Layers.
- **Non-redundancy** — No Category may duplicate the explanatory function of another.
- **Completeness** — The six Categories collectively exhaust the explanatory dimensions required for HNS.
- **Single Assignment** — A phenomenon must be interpreted through exactly one Category at a time.

8.2 Formal Definitions of the Six Categories

Category 1 — Civilization Categories

Definition: Broad abstractions that frame phenomena in terms of civilizational scale, historical context, and macro-level structures.

Explanatory Role: Provides the highest-level interpretive lens for understanding systemic or epochal phenomena.

Category 2 — Core Categories

Definition: Foundational principles, essential structures, and irreducible conceptual elements.

Explanatory Role: Identifies the fundamental logic or essence underlying a phenomenon.

Category 3 — Module Categories

Definition: Components, elements, and functional units within a system.

Explanatory Role: Breaks down phenomena into discrete, analyzable modules.

Category 4 — Application Categories

Definition: Practical uses, applied contexts, and operational implementations.

Explanatory Role: Interprets phenomena in terms of their practical function or applied purpose.

Category 5 — System Categories

Definition: Organized systems, mechanisms, and structured processes.

Explanatory Role: Explains how phenomena operate within structured systems or organized mechanisms.

Category 6 — External Categories

Definition: Extensions, outer contextual factors, and influences beyond the immediate system.

Explanatory Role: Frames phenomena in terms of external conditions, constraints, or contextual influences.

8.3 Category Ordering Principle [REVISED in v1.1]

v1.1 Revision: The ordering sequence previously referenced as a figure is now fully specified in text as normative content.

The Abstract Cognitive Categories do not form a causal hierarchy. However, for interpretive consistency, they must be read in the following conceptual order, reflecting increasing specificity and contextualization:

Position	Category	Interpretive Focus
C1 (first)	Civilization Categories	Broadest, macro-historical scale
C2	Core Categories	Foundational principles and logic
C3	Module Categories	Components and functional units
C4	Application Categories	Practical use and applied context
C5	System Categories	Organized mechanisms and processes
C6 (last)	External Categories	Outer contextual factors

This ordering is a conceptual sequence, not a causal hierarchy. Categories remain non-causal regardless of their position. The ordering exists solely to support consistent interpretive practice.

8.4 Category Assignment Rule

A phenomenon must be assigned to a Category based on its primary explanatory purpose, not its domain, scale, or effects. Example: a law may be analyzed as a Module (its components), a System (its mechanisms), or a Civilization Category (its historical role), depending on the interpretive intent.

8.5 Boundary Conditions

- **Primary Interpretation Rule** — Assign based on the primary explanatory purpose.
- **Multiple Valid Interpretations** — If multiple interpretations are possible, decompose into separate analytical units.
- **Context vs. Explanation Distinction** — External context does not automatically place a phenomenon in External Categories.
- **System vs. Module Clarification** — Systems are organized mechanisms; modules are components.
- **Civilizational Scope Exception** — A phenomenon belongs to Civilization Categories only when interpreted at macro-historical scale.

8.6 Misclassification Prevention

- **Do not treat Categories as causal** — They are explanatory, not structural.
- **Do not assign based on scale alone** — Large-scale phenomena are not automatically Civilization Categories.
- **Do not confuse System with Application** — Systems describe mechanisms; applications describe uses.
- **Do not place all contextual factors in External** — External Categories are for outer contextual structures, not incidental context.
- **Do not treat Core as 'important'** — Core refers to foundational logic, not importance.

8.7 Examples [Non-normative]

- Civilization Categories: historical eras, civilizational transitions
- Core Categories: fundamental principles, essential structures
- Module Categories: components of a law, elements of cognition
- Application Categories: practical tools, applied techniques
- System Categories: governance mechanisms, institutional processes
- External Categories: geopolitical pressures, environmental constraints

9. 36 Cells Template

The 36 Cells Template defines the structural format of each of the 36 positions created by the intersection of six Human Natural Layers (L1-L6) and six Abstract Cognitive Categories (C1-C6). This template provides the standardized structure for describing, interpreting, and referencing any cell within the HNS 36-Matrix.

9.1 Cell Identification Format

Each cell must be identified using the notation: Cell (L_i x C_j), where L_i is one of the six Human Natural Layers and C_j is one of the six Abstract Cognitive Categories. This notation is mandatory for all references within HNS.

9.2 Template Structure

Each cell must be documented using the following template:

Cell (L_i x C_j)
Layer: Name of Human Natural Layer (L_i)
Category: Name of Abstract Cognitive Category (C_j)

1. Structural Role
A formal statement describing the structural function of this intersection.
2. Interpretive Scope
The explanatory domain covered by this cell.
3. Boundary Conditions
Rules for distinguishing this cell from adjacent cells.
4. Misclassification Prevention
Guidelines to avoid common interpretive errors.
5. Non-normative Examples (Optional)
Illustrative examples that clarify the structural meaning.

9.3 Matrix-Level Constraints

- **Exhaustiveness** — All possible Layer-Category intersections must be represented.
- **Mutual Exclusivity** — No phenomenon may occupy more than one cell simultaneously.
- **Bidirectional Independence** — Layers and Categories must remain independent axes.
- **Interpretive Completeness** — The 36 cells collectively provide the full coordinate space for HNS.

10. Mapping Rules [Normative]

The Mapping Rules define the mandatory procedures for positioning any phenomenon within the HNS 36-Matrix. These rules ensure that all mappings are structurally consistent, causally accurate, interpretively coherent, and interoperable across domains.

10.1 Mapping Procedure Overview [REVISED in v1.1]

v1.1 Revision: Step 6 (Recomposition) has been added to normative mapping procedure to align with the composite phenomenon guidance described in Section 6.3.

Mapping a phenomenon into the 36-Matrix requires the following steps:

- 1. Identify the phenomenon** — Define the unit of analysis clearly.
- 2. Determine the primary causal domain** — Assign the phenomenon to exactly one Human Natural Layer (Section 7).
- 3. Determine the primary explanatory purpose** — Assign the phenomenon to exactly one Abstract Cognitive Category (Section 8).
- 4. Combine Layer and Category** — Produce the structural position Cell ($L_i \times C_j$).
- 5. Verify boundary conditions** — Ensure the mapping satisfies the constraints in Sections 7 and 8.
- 6. Document the mapping** — Use the formal notation and template defined in Section 9.
- 7. Recompose (composite phenomena only)** — If the phenomenon was decomposed into multiple components in Step 1, read the mapped cells together to understand the composite phenomenon as a structured whole. Each component retains its individual cell assignment; recomposition is an interpretive act, not a reassignment.

10.2 Layer Selection Rules

A phenomenon must be assigned to the Layer that represents its primary causal origin — not its effects, context, scale, or appearance.

- **Causal Priority Rule** — The Layer is determined by what produces the phenomenon.
- **Effect Exclusion Rule** — Effects in higher layers do not change the assigned Layer.
- **Composite Phenomena Rule** — If a phenomenon spans multiple layers, decompose it into causal components.
- **Emergent Pattern Rule** — Emergent outcomes belong to PatternOS, regardless of their origins.

10.3 Category Selection Rules

A phenomenon must be assigned to the Category that represents its primary explanatory purpose — not its domain, importance, or scale.

- **Interpretive Purpose Rule** — The Category is determined by how the phenomenon is being analyzed.
- **Non-causal Rule** — Categories must not be treated as causal domains.
- **Cross-layer Applicability Rule** — Categories apply across all Layers.
- **Context vs. Explanation Rule** — External context does not automatically place a phenomenon in External Categories.

10.4 Mapping Consistency Rules

- **Single-Cell Rule** — A phenomenon must occupy exactly one cell at a time.
- **No Dual Assignment** — A phenomenon cannot be mapped to multiple Layers or Categories simultaneously.
- **Stable Interpretation Rule** — Once mapped, the structural position remains stable unless the explanatory purpose changes.
- **Template Compliance Rule** — All mappings must follow the template in Section 9.

10.5 Boundary Handling Rules

- **Primary Domain Rule** — When in doubt, choose the Layer with the strongest causal influence.
- **Primary Explanation Rule** — When multiple Categories seem plausible, choose the one that best reflects the analytical intent.
- **Decomposition Rule** — Complex phenomena must be decomposed into simpler units before mapping.
- **Adjacency Clarification Rule** — Adjacent cells must be distinguished using boundary conditions defined in Sections 7 and 8.

10.6 Misclassification Prevention

- **Do not treat Patterns as causes** — PatternOS is emergent, not causal.
- **Do not confuse System with Application** — Systems describe mechanisms; applications describe uses.
- **Do not place all contextual factors in External** — External Categories are for outer contextual structures only.
- **Do not assign based on scale** — Large-scale phenomena are not automatically Civilization Categories.
- **Do not assign based on importance** — Core Categories are not 'important things'; they are foundational logic.

10.7 Mapping Examples [Non-normative]

Phenomenon	Layer	Category	Cell
Memory retrieval	L2 CognitiveOS	C3 Module	L2 x C3
Conversation	L3 InteractionOS	C4 Application	L3 x C4
Legal institution	L4 EnvironmentOS	C5 System	L4 x C5
Stress response	L5 LoadOS	C2 Core	L5 x C2
Cultural norms	L6 PatternOS	C1 Civilization	L6 x C1

11. Definition Statements [Normative]

This section provides the official, normative definitions for all structural components of HNS. These definitions serve as the authoritative reference for all mappings, analyses, and future extensions of HNS 1.0.

11.1 Human Natural Structure (HNS)

The complete structural model describing the causal layers, explanatory categories, and coordinate system through which human, social, and civilizational phenomena can be positioned and interpreted.

11.2 Human Natural Layers

The six causal strata that constitute the natural structure of humans, societies, and civilizations. Each layer represents a distinct causal domain and forms the vertical axis (L) of the HNS 36-Matrix.

11.3 Abstract Cognitive Categories

The six non-causal explanatory axes that provide interpretive perspectives across all Human Natural Layers. They form the horizontal axis (C) of the HNS 36-Matrix.

11.4 36-Matrix

The coordinate system formed by the intersection of the six Human Natural Layers (L1-L6) and the six Abstract Cognitive Categories (C1-C6), producing 36 unique structural positions.

11.5 Cell ($L_i \times C_j$)

A structural position within the 36-Matrix representing the intersection of one Human Natural Layer and one Abstract Cognitive Category. Each cell defines a unique interpretive domain and must follow the standardized template specified in Section 9.

11.6 Mapping

The process of assigning a phenomenon to a unique cell ($L_i \times C_j$) within the 36-Matrix by determining its primary causal domain (Layer) and primary explanatory purpose (Category), following the rules in Section 10.

11.7 Recomposition

The interpretive act of reading two or more individually mapped cells together to understand a composite phenomenon as a structured whole. Recomposition does not create a new cell assignment; each component retains its original cell. See Section 10.1, Step 7.

Afterword

The completion of the Human Natural Structure (HNS) 1.0 Foundational Specification marks the establishment of a unified coordinate system for understanding human, social, and civilizational structure.

This document does not conclude a project; it initiates a new phase in which the structure becomes available for interpretation, application, and extension.

HNS 1.0 is designed to endure. Its purpose is not to describe a moment, but to provide a stable architecture that can support future inquiry across disciplines and generations.

What This Version Establishes

Version 1.1 preserves all structural achievements of HNS 1.0 and adds three normative clarifications:

- Layer and Category ordering sequences are now fully specified in text (Sections 7.3 and 8.3).
- The Recomposition step for composite phenomena is now normatively defined in the Mapping Procedure (Section 10.1, Step 7) and Definition Statements (Section 11.7).
- Cross-references between Sections 6.3 and 10.1 are explicit, eliminating the informative/normative inconsistency present in v1.0.

To the Reader

This specification was written to provide clarity, stability, and precision. Its purpose is not to impose a worldview, but to offer a structure through which many worldviews may be understood. The completion of HNS 1.0 is not an ending — it is the moment at which the structure becomes available for others to use, test, challenge, and extend.